

Sulfasalazine for IBD in the Era of Advanced Therapy: Old is Gold

Novell Pharmaceutical Laboratories

Outlines

Inflammatory bowel disease (IBD)

Ulcerative colitis (UC)

Sulfasalazine

Jurnal Support

IBD

Inflammatory bowel disease (IBD) adalah inflamasi kronik saluran cerna dengan relaps–remisi

Dua diagnosis utama: Ulcerative colitis (UC) dan Crohn's disease (CD)

Tujuan terapi modern: kontrol gejala, mencegah komplikasi, dan mencapai penyembuhan mukosa (treat-to-target).

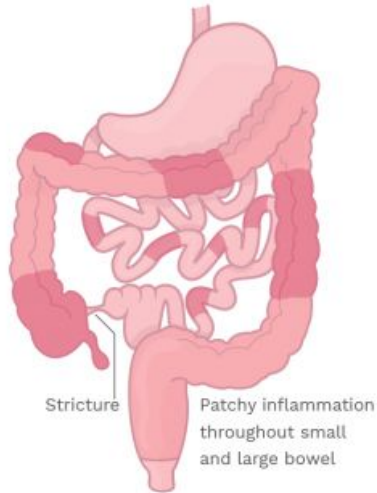
Ulcerative colitis (UC)

- Terbatas di kolon & rektum (kontinu)
- Inflamasi dominan mukosa
- Gejala utama: diare berdarah, urgensi

Crohn's disease (CD)

- Dari mulut–anus (segmental/skip)
- Transmural → fistula/striktur
- Gejala bervariasi: nyeri, diare, malabsorpsi

UC vs Crohn



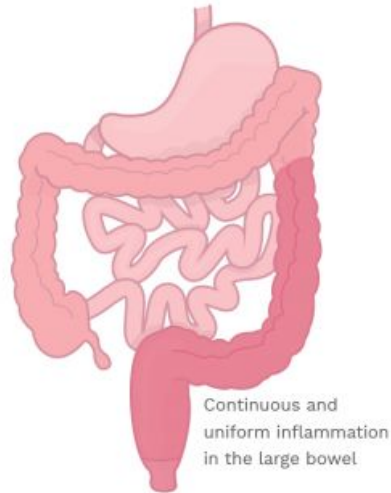
Crohn's Disease

Age of onset: 15–35 years and 55–70 years

Symptoms: Depends on location of disease. May include abdominal pain, diarrhea, weight loss and fatigue.

Bloody stool: Variable

Malnutrition: Common



Ulcerative Colitis

Age of onset: 15–35 years and 55–70 years

Symptoms: May include stool urgency, fatigue, increased bowel movements, mucous in stool, nocturnal bowel movements and abdominal pain.

Bloody stool: Common

Malnutrition: Less common

UC: inflamasi kontinu dari rektum ke proksimal, terbatas pada kolon, perdarahan rektal dominan

Crohn: lesi segmental/skip, dapat melibatkan ileum & kolon, transmural

→ fistula/striktur, juga lebih sering Lokasi/ekstensitas menentukan pilihan obat: terapi kolon-targeted lebih relevan bila kolon dominan

Crohn's Classification

Table 1 Vienna and Montreal classification for Crohn's disease

	Vienna	Montreal
Age at diagnosis	A1 below 40 y A2 above 40 y	A1 below 16 y A2 between 17 and 40 y A3 above 40 y
Location	L1 ileal L2 colonic L3 ileocolonic L4 upper	L1 ileal L2 colonic L3 ileocolonic L4 isolated upper disease*
Behaviour	B1 non-stricturing, non-penetrating B2 stricturing B3 penetrating	B1 non-stricturing, non-penetrating B2 stricturing B3 penetrating p perianal disease modifier†

*L4 is a modifier that can be added to L1–L3 when concomitant upper gastrointestinal disease is present.

†“p” is added to B1–B3 when concomitant perianal disease is present.

UC's Classification

Table 2 Montreal classification of extent of ulcerative colitis (UC)

Extent		Anatomy
E1	Ulcerative proctitis	Involvement limited to the rectum (that is, proximal extent of inflammation is distal to the rectosigmoid junction)
E2	Left sided UC (distal UC)	Involvement limited to a proportion of the colorectum distal to the splenic flexure
E3	Extensive UC (pancolitis)	Involvement extends proximal to the splenic flexure

Table 3 Montreal classification of severity of ulcerative colitis (UC)

Severity		Definition
S0	Clinical remission	Asymptomatic
S1	Mild UC	Passage of four or fewer stools/day (with or without blood), absence of any systemic illness, and normal inflammatory markers (ESR)
S2	Moderate UC	Passage of more than four stools per day but with minimal signs of systemic toxicity
S3	Severe UC	Passage of at least six bloody stools daily, pulse rate of at least 90 beats per minute, temperature of at least 37.5°C, haemoglobin of less than 10.5 g/100 ml, and ESR of at least 30 mm/h

ESR, erythrocyte sedimentation rate.

UC's Classification

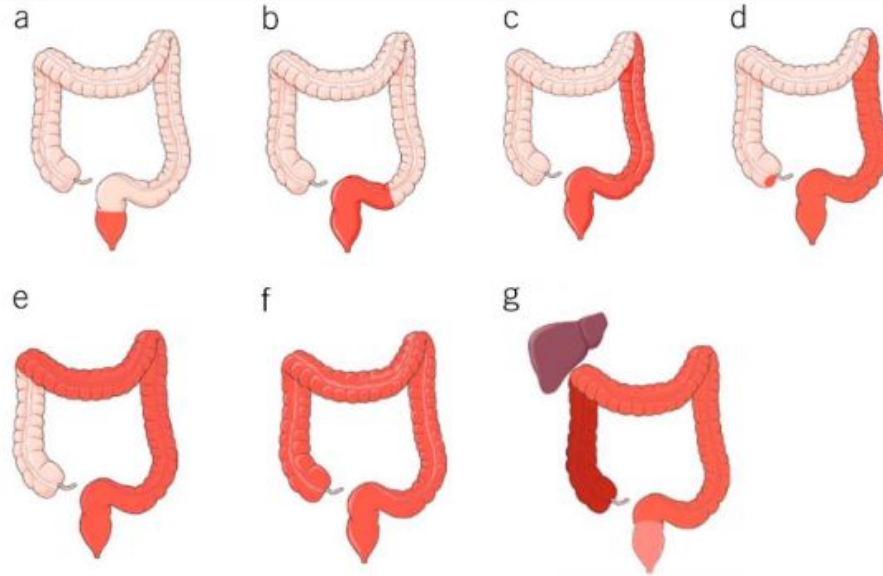
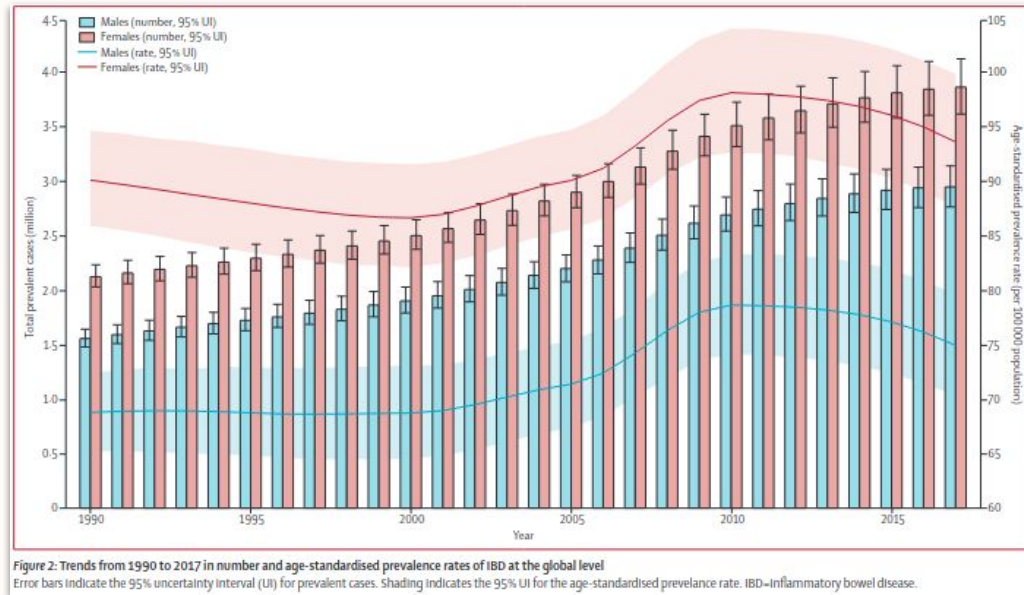


Figure 1. Different clinical phenotypes described with ulcerative colitis. Isolated proctitis (a), proctosigmoiditis (b), left-sided colitis (c), extensive colitis (d), and pancolitis (e) are the traditional described phenotypes based on extent of mucosal involvement (historically by barium radiography or endoscopy, more recently defined by histology as well). The primary sclerosing cholangitis phenotype (f) of relative or absolute rectal sparing can be seen in patients and considered a variant of the traditional extent-based phenotypes but is medically managed similarly. The periappendiceal patch or cecal patch phenotype (g) is sometimes seen in patients with limited distal colitis and is similarly managed as well.

Epidemiologi IBD



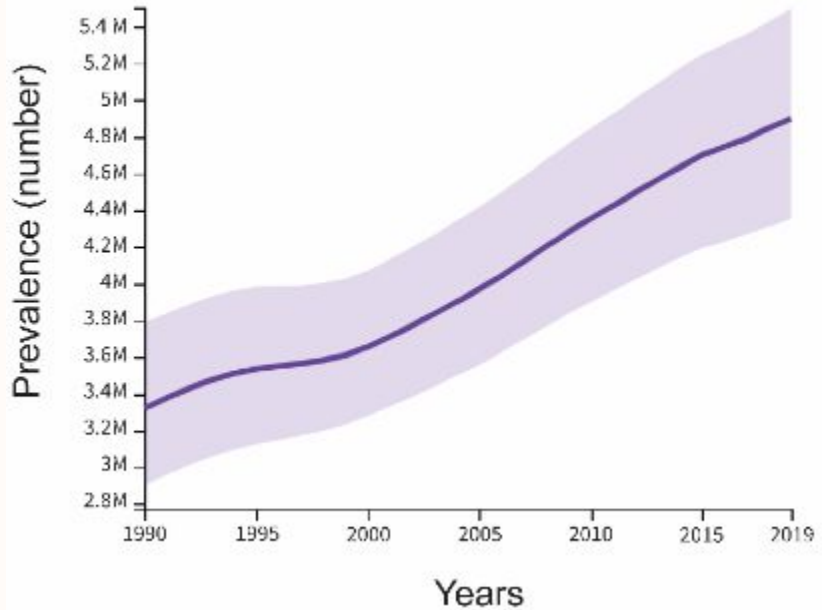
6,8 juta

GBD 2017 memperkirakan ~6,8 juta orang hidup dengan IBD pada 2017

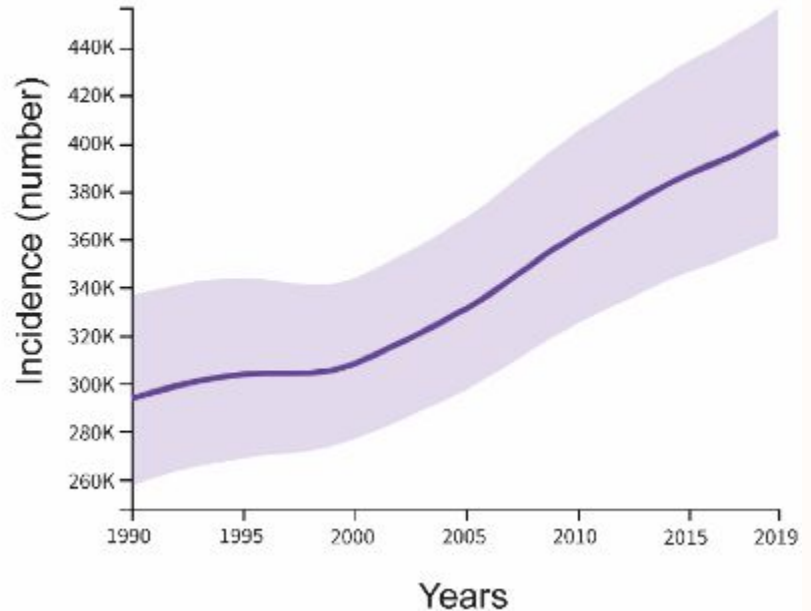
Beban meningkat di banyak negara seiring perubahan lingkungan, urbanisasi, dan diagnosis yang membaik

IBD Prevalensi & Insidens

IBD prevalence



IBD incidence



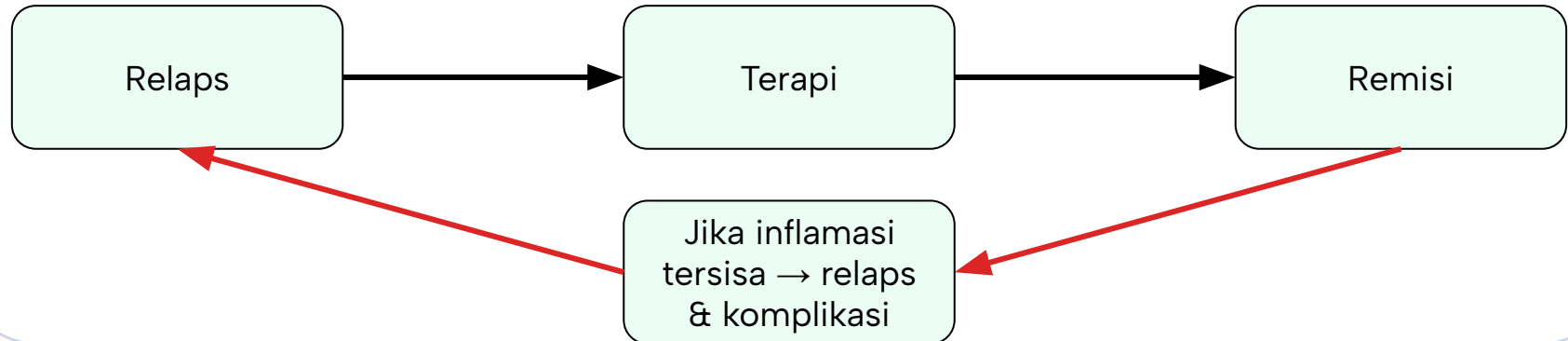
Natural history: Relaps+remisi

IBD umumnya bersifat kronik dengan fase relaps dan remisi → inflamasi yang tidak terkontrol memicu komplikasi

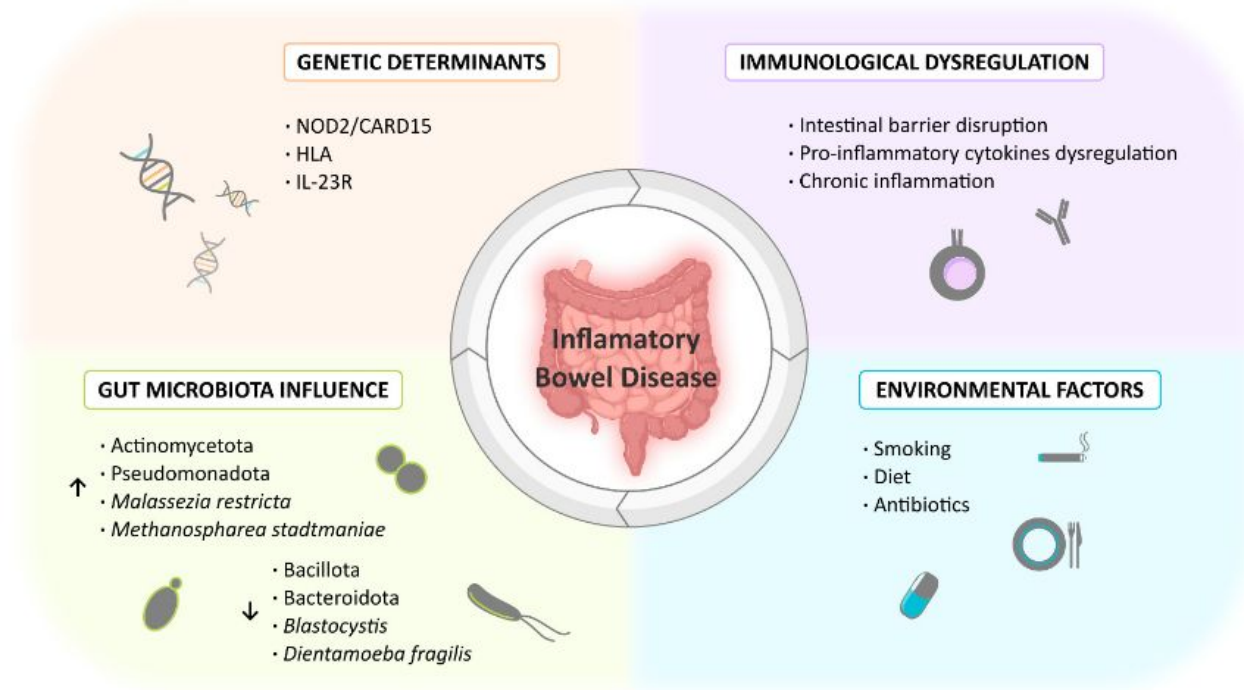
Konsekuensi: Kebutuhan steroid berulang, hospitalisasi, komplikasi struktural (khususnya CD), dan pembedahan

Butuh strategi modern menargetkan kontrol inflamasi objektif (biomarker/endoskopi) selain gejala

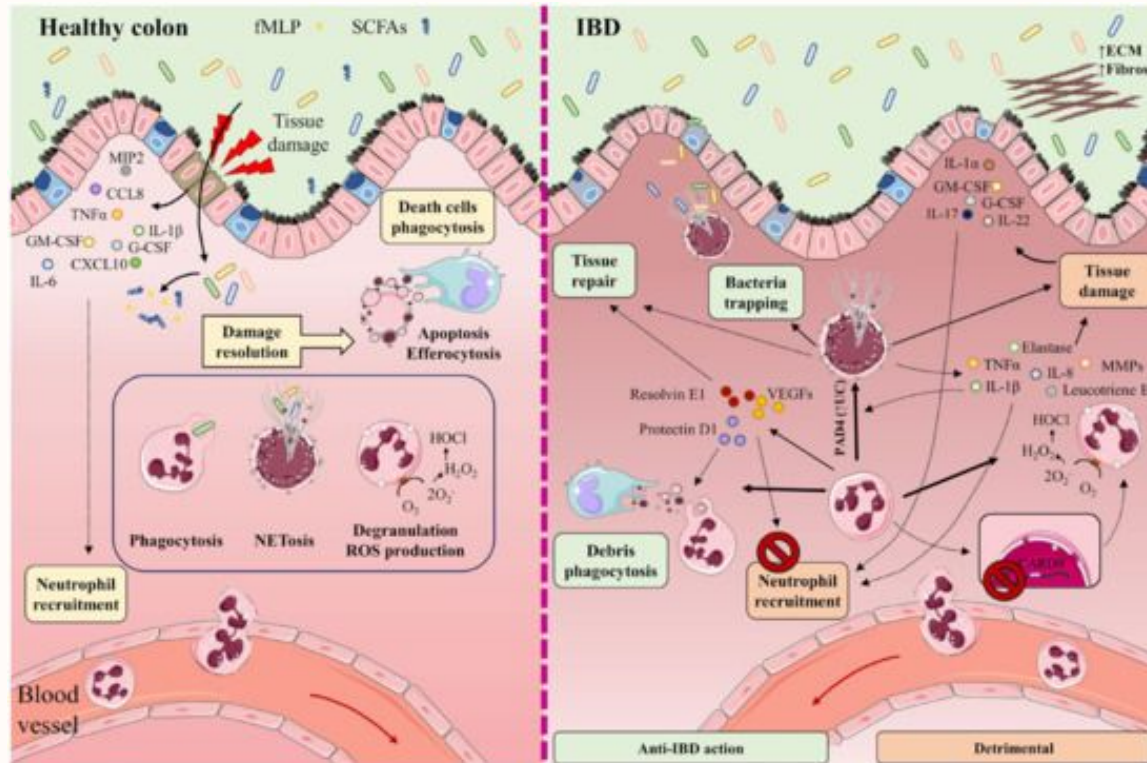
Siklus klinis



Faktor Risiko IBD

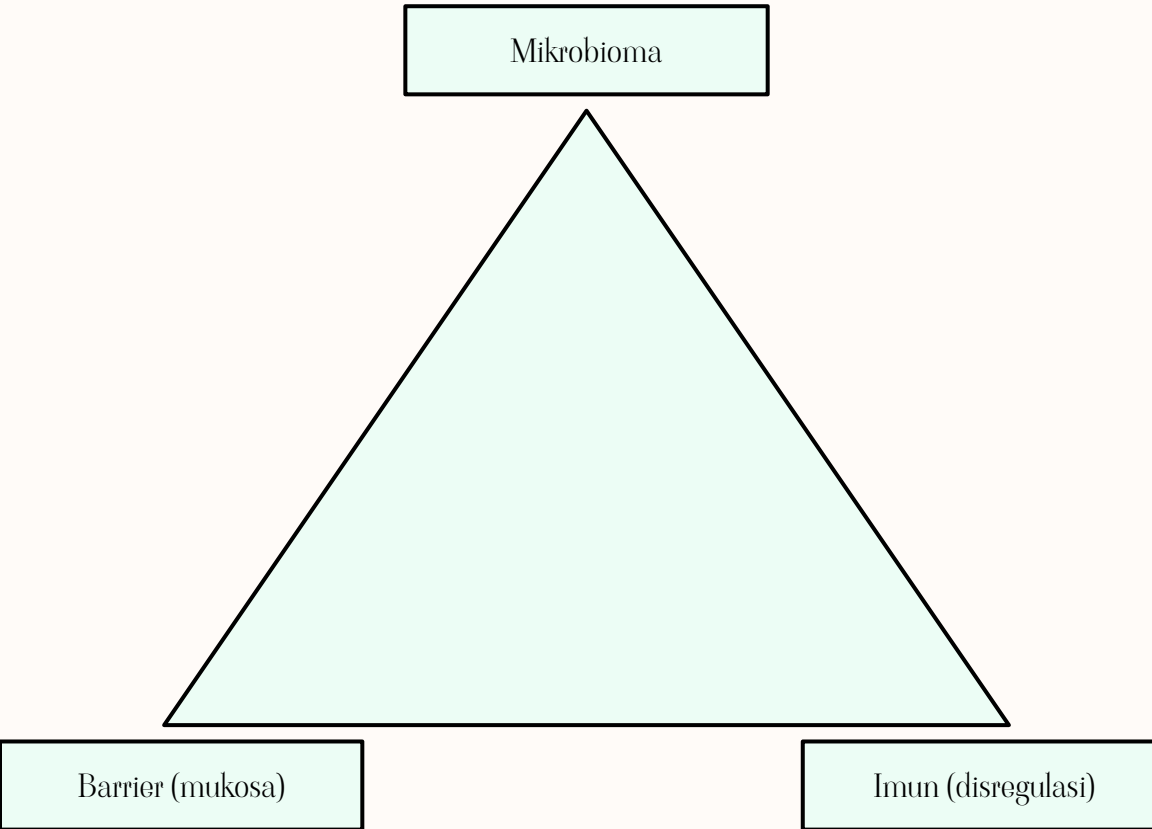


Patofisiologi IBD



1. Barrier usus rusak $\rightarrow$ sinyal inflamasi + sinyal bakteri menarik neutrofil dari darah ke mukosa.
2. Neutrofil datang untuk “bunuh kuman”: fagositosis, degranulasi, ROS, dan NETs.
3. Normalnya selesai lalu padam: neutrofil apoptosis + dibersihkan (efferocytosis) $\rightarrow$ resolusi & perbaikan mukosa.
4. Di IBD, neutrofil/NETs jadi “dua sisi”: bisa membantu wound healing (VEGF, resolvin/protectin) & menjebak bakteri.
5. Tapi jika berlebihan, malah merusak: elastase, MMPs, IL-8/TNF- α /IL-1 β , LTB₄, ROS $\rightarrow$ barrier makin bocor.
6. Terbentuk lingkaran setan: barrier bocor $\rightarrow$ rekrutmen neutrofil makin banyak $\rightarrow$ NETs/ROS naik $\rightarrow$ inflamasi makin berat.

Patofisiologi IBD



1. kerentanan genetik + paparan lingkungan dapat mengganggu barrier mukosa dan respons imun.
2. Disbiosis & interaksi mikrobioma memicu aktivasi imun innate/adaptif → inflamasi kronik
3. Terapi anti-inflamasi (termasuk 5-ASA/SSZ) terutama bekerja pada inflamasi mukosa kolon

Diagnosis UC

Diagnosis, assessment, monitoring, and prognosis of ulcerative colitis

1. The diagnosis of UC should be suspected in patients with hematochezia, increased stool frequency, or bowel urgency
2. Infectious etiologies should be excluded at the time of diagnosis
3. Colonoscopy with intubation of the ileum and biopsies of affected and unaffected areas should be obtained to confirm the diagnosis of UC, with mucosal biopsies interpreted by a pathologist, preferably one with expertise in gastrointestinal pathology
4. Categories of disease extent include (i) proctitis (within 18 cm of anal verge, distal to rectosigmoid junction), (ii) left-sided colitis (extending from sigmoid to splenic flexure), (iii) extensive colitis (beyond splenic flexure which includes those with involvement of the entire colorectum [pancolitis])
5. If the terminal ileum is normal, further evaluation of the stomach and small bowel by upper endoscopy and cross-sectional imaging is not needed unless there are other symptoms or findings to suggest proximal gastrointestinal involvement or a diagnosis of Crohn's disease rather than UC
6. Definitions of disease severity are needed to guide treatment decisions; definitions should be based on (i) patient-reported outcomes (bleeding, normalization of bowel habits, bowel urgency), (ii) the inflammatory burden (endoscopic assessment including extent and severity and markers of inflammation including FC, CRP, and serum albumin), (iii) disease course (need for hospitalization, need for steroids, failure to respond to medications), and (iv) disease impact (HRQoL and social functioning)
7. Endoscopic severity should be reported using a validated endoscopic scale such as the Mayo Endoscopic Score or the UC Endoscopic Index of Severity
8. Disease assessment and monitoring in response to therapy and during maintenance and periods of suspected relapse may be performed with FC, CRP, endoscopic assessment with flexible sigmoidoscopy or colonoscopy, and/or intestinal ultrasound

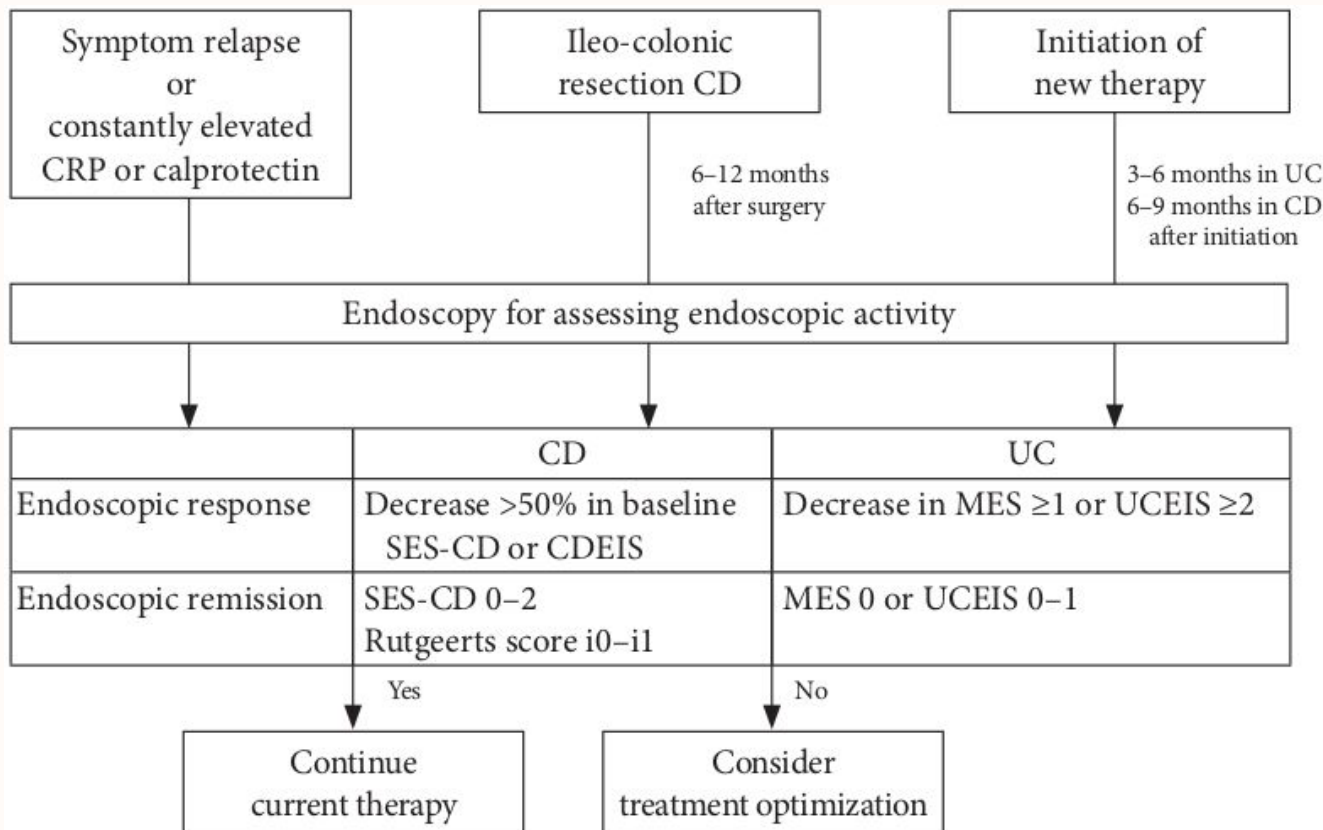
Endoscopic Scoring System

Table 1. Endoscopic scoring systems commonly used in ulcerative colitis and Crohn's disease

Scoring system		Descriptors	Score range	Segment consideration	Inter-observer agreement (κ score)	Interpretation	Strength	Weakness
UC	MES	Erythema, vascularity, friability, bleeding, erosions, ulceration	0. Normal or inactive disease 1. Decreased vascularity, mild friability, erythema 2. Marked erythema, absence of vascularity, friability, erosion 3. Ulcers and spontaneous bleeding	Most severe finding	0.53 Inexperienced 0.71 Experienced	0. Normal 1. Mild 2. Moderate 3. Severe	1. Widely used in clinical trials 2. Easy to use	1. No consideration of distribution and extend 2. Overlap between scores in some descriptors
	UCEIS	Vascularity Bleeding Erosions/ulcerations	1. Vascular pattern: 0-2, bleeding and ulceration: 0-3 2. Total score: 0-8	Most severe area	0.5 Experienced	Not validated, But usually Remission ≤1 Mild 2-4 Moderate 5-6 Severe 7-8	1. Only validated index in UC 2. Easy to assess 3. Better score range than MES	1. No validated severity definition 2. No consideration regarding change and healing
CD	CDEIS	1. Superficial ulcer 2. Deep ulcer 3. Surface affected 4. Ulcerated surface	Total score: 0-44	5. Terminal ileum, ascending, transverse, descending and sigmoid colon, rectum	0.67 Inexperienced 0.83 Experienced	Not validated, but usually Healed: ≤3 Mild: <5 Moderate 5-15 Severe: >15	1. Consider segments that are not visualized 2. Good correlation with disease surface 3. Prognostic relevance	1. Difficult to assess size of surface, ulcer and ulcer depth 2. Difficult to calculate
	SES-CD	1. Ulcer size (aphthous <0.5 cm, large 0.5-2 cm, very large >2 cm) 2. Surface ulcerated 3. Surface affected 4. Stricture	1. Total: 0-56 (total of all segments) 2. Value of each descriptors: 0-3	5. Terminal ileum, ascending, transverse, descending, and sigmoid colon, rectum	0.68 Inexperienced 0.93 Experienced	Not validated but usually Healed: 0-2, Mild: 3-6, Moderate: 7-15 Severe: ≥16	1. Simpler than CDEIS 2. Clear definition of score 3. Prognostic relevance	1. Underestimation in case only 1 segment is involved 2. No consideration of uninvolved segments
Post-op CD	Rutgeerts score	1. Aphthous ulcers (≤5 and >5) 2. Large ulcers 3. Inflamed intervening mucosa 4. Diffuse inflammation	Score of each descriptor, 0 to 34 (in modified RS, it subdivided into 12a, 12b)	Neoterminal ileum examined	0.57 Inexperienced 0.71 Experienced	Validated Low risk: 0-1 Medium risk: 2 High risk: 3-4	1. Gold standard in post-op setting 2. Validated cut off 3. Easy to use	1. Not fully validate 2. Only available in prediction of post-op recurrence

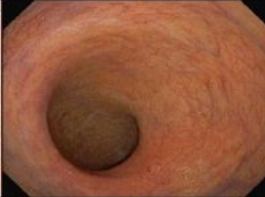
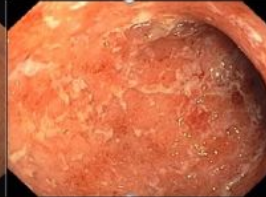
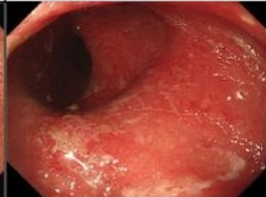
UC, ulcerative colitis; CD, Crohn's disease; MES, Mayo endoscopic sub-score; UCEIS, Ulcerative Colitis Endoscopic Index of Severity; CDEIS, Crohn's Disease Endoscopic Index of Severity; SES-CD, Simple Endoscopic Score for Crohn's Disease; RS, Rutgeerts score.

Suggested Algorithm for Endoscopic Assessment



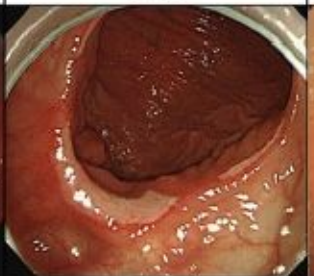
Endoscopic Scoring System

Endoscopic features of each descriptor in ulcerative colitis endoscopic index of severity

Score	0	1	2	3
Vascular pattern	 Normal	 Patchy obliteration	 Obliterated	
Bleeding	 No visible blood	 Mucosal	 Luminal mild	 Luminal moderate
Erosion and ulcers	 No visible erosion	 Erosions	 Superficial ulcer	 Deep ulcer

Endoscopic & Rutgeerts Score

Rutgeerts score for postoperative Crohn's disease (i2a and i2b by modified Rutgeerts score)

i0	i1	i2		i3	i4
		i2a	i2b		
					
No lesion in distal ileum	≤5 Aphthous lesions	Lesions confined to ileocolonic anastomosis	>5 Aphthous lesions with normal mucosa between the lesions	Diffuse aphthous ileitis with diffusely inflamed mucosa	Diffuse inflammation with already large ulcers and/or narrowing

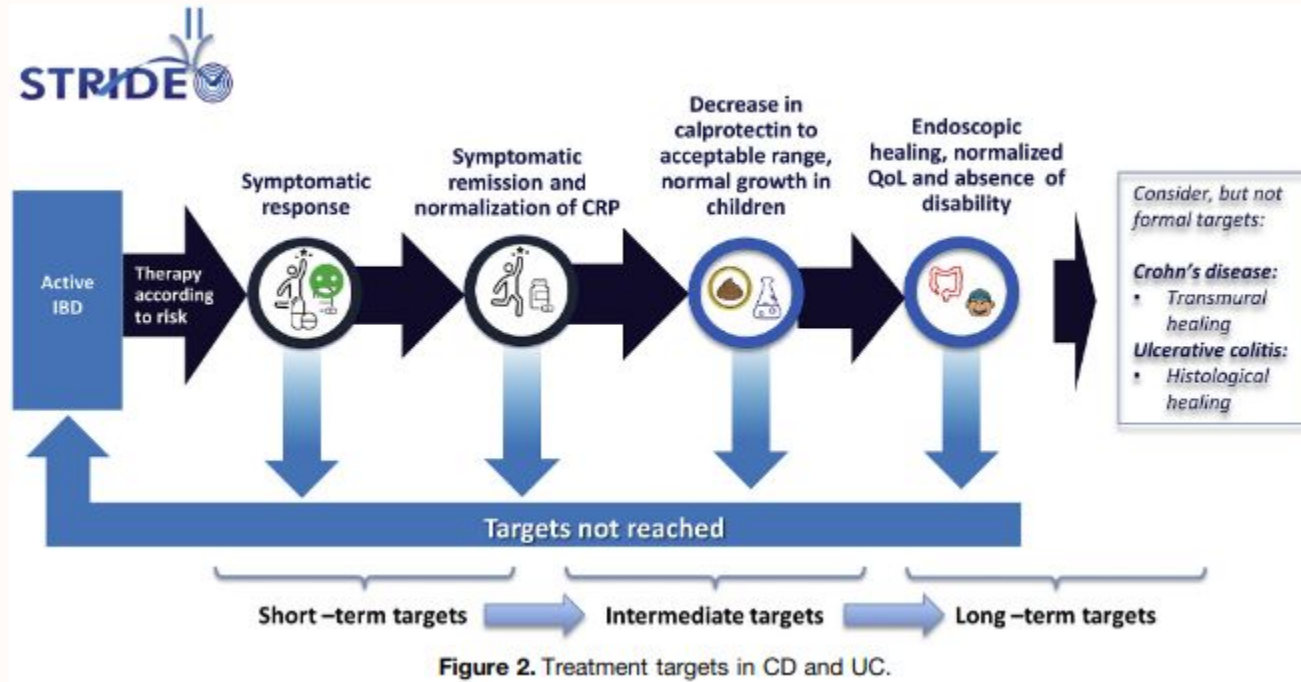
UC

Gejala:
Diare berdarah
Urgensi
Tenesmus

Komplikasi:
Anemia
Hospitalisasi flare
Steroid exposure

Target Terapi:
Remisi klinis +
endoskopi
(treat-to-target)

Treatment Goals in IBD



Pemantauan & Evaluasi Terapi

Minggu 2–4:
Cek early response →
frekuensi BAB, darah,
urgensi, nyeri
+ tren CRP dan/atau fecal
calprotectin

Minggu 6–8:
Target clinical remission
(kalau belum: cek adherence,
optimasi dosis,
pertimbangkan eskalasi)

3–6 bulan
Target endoscopic
improvement/healing
(terutama bila biomarker
tetap tinggi atau gejala
menetap)

IBD Therapy: Dari Konvensional Sampai Advanced

Konvensional:
5-ASA
Budesonide
systemic steroid

Immunomodulator:
Thiopurine
MTX (lebih ke
maintenance/selec
ted)

Biologics:
Anti-TNF
Anti-integrin
Anti-IL12/23
Anti-IL23

Small molecules:
JAK inhibitor
S1P modulator

Medication in IBD

Table 4. Time (Mean Number of Weeks) Required for Achieving the Goal After Starting Treatment for CD (n = 39) and UC (n = 36), Based on the Delphi-like Process and the Systematic Review of the Evidence

	Clinical response	Clinical remission	Norm of CRP/ESR	Decrease of FC ^a	EH
Crohn's disease					
Oral steroids/EEN	2	4	5	8	13
Budesonide	3	6	8	10	15
Thiopurines	11	15	15	17	24
Methotrexate	9	14	14	15	24
Anti-TNF	2-4	4-6	9	11	17
Vedolizumab	11	17	15	17	24
Ustekinumab	7	13	11	14	19
Ulcerative colitis					
Oral 5-ASA	4	8	8	10	13
Oral Steroids	2	2	5	8	11
Locally active steroids ^b	3	8	8	9	13
Thiopurines	11	15	15	15	20
Adalimumab	6	11	10	12	14
Infliximab	5	10	9	11	13
Vedolizumab	9	14	14	15	18
Tofacitinib	6	11	9	11	14

NOTE. Given the paucity of high-quality scientific data, the data in this table should be considered merely as a rough estimate of experts' opinion.

5ASA, 5-aminosalicylic acid; EEN, exclusive enteral nutrition.

^aBelow a desired threshold outlined in the text and [Supplementary Table 4](#).

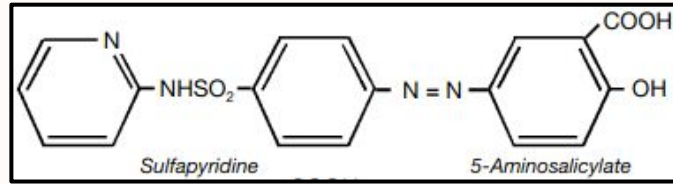
^bBeclomethasone Dipropionate (BDP), Budesonide MMX.

6. In patients with mildly to moderately active ulcerative proctitis, we recommend rectal 5-aminosalicylate acid (5-ASA) therapies at a dose of 1 g/daily for induction of remission (Strong recommendation, moderate quality evidence).
7. For patients with mildly to moderately active proctitis not responsive to topical 5-ASA, we suggest tacrolimus suppository or beclomethasone suppository over no treatment (Conditional recommendation, low quality of evidence).
8. For patients with mildly to moderately active proctitis or left-sided colitis, we suggest use of topical corticosteroids (suppository, foam, enema), over no treatment (Conditional recommendation, very low quality of evidence).
9. In patients with mildly to moderately active proctitis or left-sided colitis, we recommend rectal 5-ASA enemas at a dose of at least 1 g/daily preferred over rectal steroids for induction of remission (Strong recommendation, moderate quality of evidence).
10. In patients with mildly to moderately active left-sided UC, we suggest rectal 5-ASA enemas at a dose of at least 1 g/daily combined with oral 5-ASA at a dose of at least 2.0 g/daily compared with oral 5-ASA therapy alone for induction of remission (Conditional recommendation, low quality of evidence).
11. In patients with mildly to moderately active left-sided UC, who are intolerant or nonresponsive to oral and rectal 5-ASA at appropriate doses (oral at least 2.0 g daily and rectal at least 1 g daily), we recommend oral budesonide Multi Matrix System (MMX) 9 mg/d for induction of remission (Strong recommendation, moderate quality of evidence).
12. In patients with mildly to moderately active extensive colitis, oral 5-ASA at a dose of at least 2.0 g daily is recommended to induce remission (Strong recommendation, moderate quality of evidence).
13. In patients with UC of any extent who fail to respond to 5-ASA therapy, we recommend oral systemic corticosteroids to induce remission (Strong recommendation, low quality of evidence).
14. In patients with mildly to moderately active UC who fail to reach remission with appropriately dosed 5-ASA (at least 2–4.8 g daily oral mesalamine and/or at least 1 g daily rectal mesalamine), we suggest against changing to an alternate 5-ASA formulation to induce remission. Alternative therapeutic classes should be considered (Conditional recommendation, low quality of evidence).
15. In patients with mildly active UC of any extent, we suggest using a low dose (2.0–2.4 g) of 5-ASA, in comparison with a higher dose (4.8 g), because there is no difference in remission rate (Conditional recommendation, very low quality of evidence).
16. In patients with mildly to moderately active UC of any extent not responding to oral 5-ASA, we recommend the addition of budesonide MMX 9 mg/d to induce remission (Strong recommendation, moderate quality of evidence).
17. In patients with mildly to moderately active UC of any extent using 5-ASA to induce remission, we recommend either once daily or more frequently dosed oral 5-ASA based on patient preference to optimize adherence because efficacy and safety are no different (Strong recommendation, moderate quality evidence).
18. In patients with mildly active ulcerative proctitis, we recommend rectal 5-ASA at a dose of 1 g daily for maintenance of remission (Strong recommendation, moderate quality of evidence).
19. In patients with mildly active left-sided or extensive UC, we recommend oral 5-ASA therapy (at least 1.5 g/d) for maintenance of remission (Strong recommendation, moderate quality of evidence).
20. We recommend against systemic, budesonide MMX, or topical corticosteroids for maintenance of remission in patients with UC (Strong recommendation, moderate quality of evidence).

Rubin DT, Ananthakrishnan AN, Siegel CA, Barnes EL, Long MD. ACG Clinical Guideline Update: Ulcerative Colitis in Adults. Am J Gastroenterol. 2025;120(6):1187-1224. Published 2025 Jun 3. doi:10.14309/ajg.0000000000003463

Terapi UC (2025): Mild+Moderate

Mild–moderate UC: 5–ASA oral/topikal adalah terapi lini awal
Sulfasalazine (SSZ) adalah opsi 5–ASA klasik terutama untuk kebutuhan cost/akses dan tidak ada kontraindikasi sulfa



Sulfasalazine

SSZ adalah 5-ASA klasik yang telah lama dipakai pada UC dan sebagian Crohn kolonik

SSZ adalah prodrug yang membawa 5-ASA ke kolon melalui ikatan azo yang dipecah bakteri kolon

Pemecahan SSZ menghasilkan 5-ASA (aksi lokal mukosa) dan sulfapyridine

Metabolisme SSZ

SSZ
(oral)

Azo-reduksi
bakteri kolon

5-ASA (lokal)
+
Sulfapyridine (sistemik)

Kapan Memulai Terapi Sulfasalazin?

Kolitis Ulseratif (UC) ringan–sedang

SSZ digunakan sebagai terapi awal (5-ASA)

Pertimbangan harga lebih ekonomis

Pemeliharaan Remisi

mempertahankan remisi UC, pertimbangkan tolerabilitas pasien terlebih dahulu

Penyakit Crohn

Kurang direkomendasikan rutin

Tapi dapat digunakan pada Crohn ringan walaupun terbatas di kolon

Colonic Crohn's mild (bukan ileal/perianal kompleks)

Dosis Sulfasalazine

Dosis Induksi

- 6–8 kaplet 500 mg/hari = 3–4 g/hari
- Dibagi 3–4 dosis

Dosis Pemeliharaan

- 4 kaplet 500 mg/hari = 2 g/hari
- BID–QID sesuai toleransi

Pediatrik >6 tahun:

Induksi: 40–60 mg/kgBB/hari dibagi 3–6 dosis

Maintenance: 30 mg/kgBB/hari dibagi 4 dosis

Populasi <2 tahun: tidak direkomendasikan

Sulfasalazine for the treatment of CD

Kategori	Sulfasalazine
Dosis	3.2 – 4 g/hari
Indikasi	Penyakit aktif tingkat ringan hingga sedang, pemeliharaan pasca operasi
Efek Samping Penting	Sakit kepala, mual, dan nyeri perut (sering terjadi pada penggunaan sulfasalazine, hingga 45% pasien); trombopenia; nefritis interstitial; pankreatitis.
Pemantauan	Fungsi hati, hitung darah lengkap (full blood count), dan terutama fungsi ginjal.

Sejarah Sulfasalazine

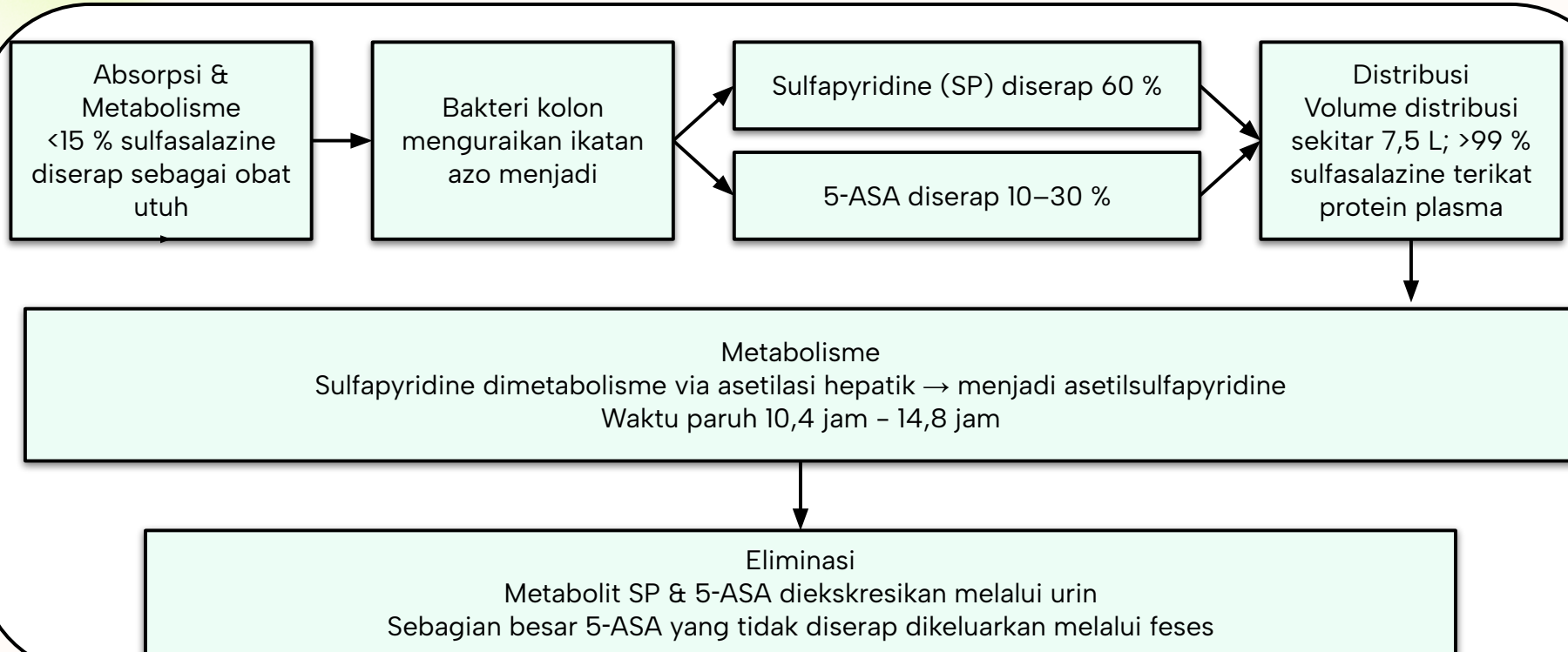
Pengembangan awal (1940-an)
Sulfasalazine pertama kali dibuat pada tahun 1940 oleh Nanna Svartz di Swedia.
Dirancang untuk mengobati artritis reumatoid

Percobaan terkontrol pertama dilakukan pada tahun 1962 → diikuti percobaan pemeliharaan pada tahun 1973 untuk Ulseratif Kolitif



Penemuan manfaat terapi berasal dari komponen 5-ASA yang dilepaskan di kolon, sedangkan sulfapyridine menyebabkan banyak efek samping

Farmakokinetik Sulfasalazine



Sulfasalazine & Its Metabolites in UC & CD

Table 2. Therapeutic Efficacy of the Three Regimens after Six Weeks of Treatment.

VARIABLE	SULFA-SALAZINE	SULFA-PYRIDINE	5-AMINO-SALICYLIC ACID
Number of patients	11	7	14
Activity index *			
Before treatment	245 ± 129	157 ± 24	251 ± 65
After treatment	100 ± 71 †	139 ± 17 ‡	90 ± 93 §
Improvement (%) *¶	57 ± 25	13 ± 6	66 ± 27
Remissions	7 (64)	1 (14)	12 (86)
Failure of remission in patients with Crohn's disease (%)	50	100	50
Improvement in stool quality	9 (82)	1 (14)	11 (79)
Failure of stool improvement in patients with Crohn's disease (%)	50	100	50

*Mean ± S.D.

†P<0.001 by paired Student's t-test.

‡Not significant by paired Student's t-test.

§P<0.0001 by paired Student's t-test.

¶The unpaired Student's t-test was used. P = 0.0003 for the comparison between sulfasalazine and sulfapyridine; P = 0.0001 between sulfapyridine and 5-aminosalicylic acid. The difference between sulfasalazine and 5-aminosalicylic acid was not significant.

||Figures in parentheses denote percentages. The chi-square test was used for comparisons. P = 0.01 for comparisons between remission rates for sulfapyridine and 5-aminosalicylic acid, between stool-improvement rates for sulfasalazine and sulfapyridine, and between stool-improvement rates for sulfapyridine and 5-aminosalicylic acid. All other differences were not significant.

Metode:

- Randomized controlled trial

Populasi:

- 32 pasien IBD: 9 Crohn's disease + 23 ulcerative colitis

Intervensi:

- Sulfasalazine 1 g 3x/hari selama 6 minggu (n=11)
- Sulfapyridine 0,5 g 3x/hari selama 6 minggu (n=7)
- 5-aminosalicylic acid (5-ASA) 0,5 g 3x/hari (bentuk suppositoria) selama 6 minggu (n=14)

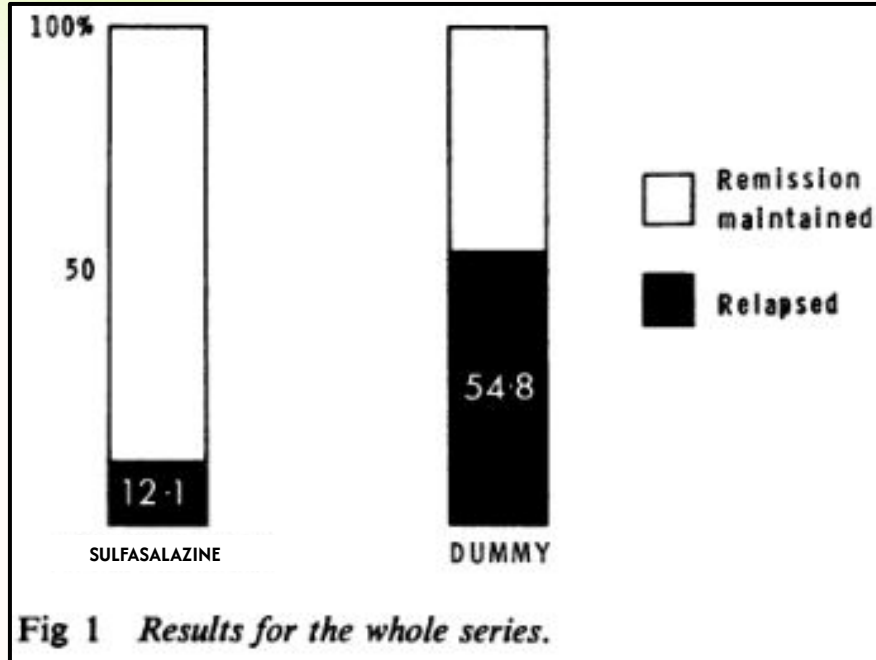
Hasil:

- Activity index membaik signifikan pada sulfasalazine ($\approx 245 \pm 129 \rightarrow 100 \pm 71$; $P < 0,001$) dan 5-ASA ($\approx 251 \pm 65 \rightarrow 90 \pm 93$; $P < 0,0001$)
- Kualitas feses membaik: sulfasalazine 82%, 5-ASA 79%, sulfapyridine 14%
- Remisi klinis: 5-ASA 86%, sulfasalazine 64%, sulfapyridine 14%

Kesimpulan:

- Sulfasalazine DAN 5-ASA secara signifikan efektif memperbaiki activity index, kualitas feses dan remisi klinis

Sulphasalazine in UC



Metode:

- Uji klinis terkontrol (dummy tablet), randomized (maintenance trial)

Populasi:

- 64 pasien ulcerative colitis (UC) yang sudah remisi (bebas gejala + tidak ada inflamasi pada sigmoidoskopi & biopsi), sebelumnya min. 1 tahun memakai sulfasalazine maintenance

Intervensi:

- Sulfasalazine 0,5 g 4x/hari (total 2 g/hari) selama 6 bulan vs dummy tablets

Hasil:

- Relaps 4/33 (12,1%) pada sulfasalazine vs 17/31 (54,8%) pada dummy; $\chi^2 = 13,2$; $P < 0,001$

Kesimpulan:

- Sulfasalazine maintenance secara signifikan menurunkan angka relaps UC dibanding dummy sehingga layak diteruskan jangka panjang bila tidak ada efek samping bermakna.

Sulphasalazine vs Mesalazine

TABLE III—Demographic and pretrial characteristics of patients studied. Intention to treat analysis

	Treatment group		Significance
	Coated mesalazine (n = 115)	Sulphasalazine (n = 105)	
No of men/No of women	71/44	61/44	0.365*
Age (years):			
Mean (SD)	38.7 (12.9)	39.5 (14.5)	0.6807†
Range	18-70	18-69	
Duration of ulcerative colitis (years):			
Mean (SD)	4.4 (5.7)	5.6 (6.4)	0.2029‡
Range	0-30	0-30	
Extent of disease (No (%) of patients):			
Rectum/sigmoid	32 (28)	26 (25)	0.221*
Partial colon	42 (37)	38 (36)	
Total colon	5 (4)	9 (9)	
Unspecified	36 (31)	32 (30)	
Disease description (No (%) of patients):			
Continuous	56 (49)	43 (41)	0.062*
Episodic	57 (50)	59 (56)	
Unspecified	2 (2)	3 (3)	
Clinical activity index:			
Mean (SD)	7.8 (2.3)	8.0 (2.5)	0.8604‡
Range	3-17	5-15	
Endoscopic index:			
No studied	107	95	0.4975‡
Mean (SD)	8.5 (2.1)	8.7 (2.2)	
Range	4-12	4-12	

*Cochran-Mantel-Haenszel statistics.

†t Test.

‡Wilcoxon two-sample test.

Metode:

- RCT

Populasi:

- 164 pasien dewasa dengan kolitis ulserativa aktif ringan hingga sedang

Intervensi:

- Mesalazine enteric-coated 1.5 g/hari vs sulfasalazine 3 g/hari selama 8 minggu

Hasil:

- Pada 4 minggu, remisi klinis tercapai pada 71% pasien mesalazine dan 66% sulfasalazine
- Setelah 8 minggu 74% vs 81%. Remisi endoskopik hampir sama (49% vs 47%)
- Efek samping lebih sering terjadi pada sulfasalazine (24%) dibanding mesalazine (14%)

Kesimpulan:

- Sulfasalazine memiliki efikasi yang setara dengan mesalazine untuk kolitis ulserativa

Sulphasalazine vs Mesalazine

Fig. 3 Individual and summary relative risk for the outcome of "relapse" for studies including comparison of mesalamine and sulfasalazine therapy in management of UC

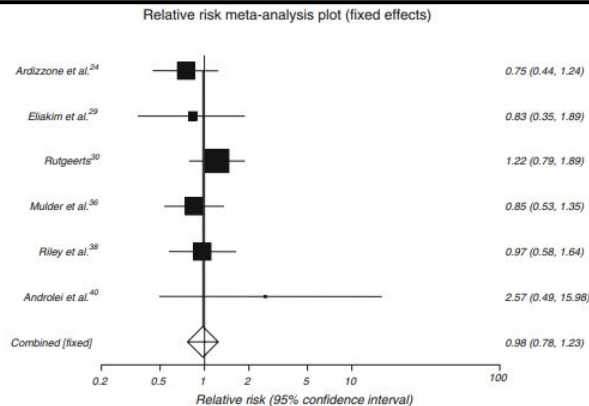
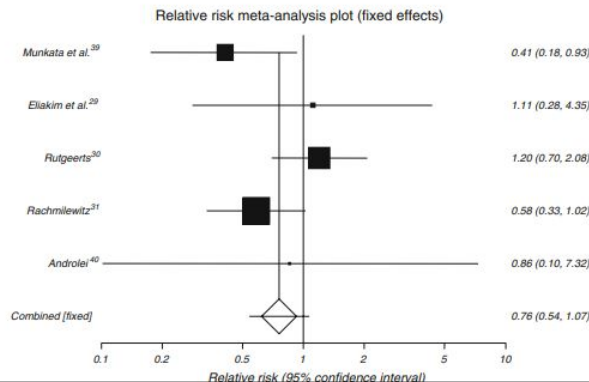


Fig. 4 Individual and summary relative risk for the outcome of "any adverse events" for studies including comparison of mesalamine and sulfasalazine therapy in the management of UC



Metode:

- Meta-analisis RCT (search 1966–Apr 2008)

Populasi:

- Pasien ulcerative colitis (mild–moderate); total 20 RCT

Intervensi:

- Sulfasalazine (SSZ) dibandingkan 5-ASA (mesalamine/olsalazine/balsalazide) untuk induksi perbaikan & maintenance remisi

Hasil:

- SSZ vs mesalamine: overall improvement RR 1,04 (95% CI 0,89–1,21; P=0,63); relapse RR 0,98 (0,78–1,23; P=0,85)
- SSZ vs olsalazine: withdrawals karena efek samping lebih rendah pada SSZ (RR 0,30; 0,12–0,77; P=0,01)

Kesimpulan:

- sulfasalazine memiliki efikasi yang mirip dengan 5-ASA untuk induksi dan maintenance UC

Old is Gold

Value:
Biaya & Akses

Validated:
Efektivitas terbukti (data lama, tapi konsisten)

Versatile:
Cocok untuk subset (kolon + beberapa EIM)

Take Home Messages

Old is Gold

Di era biologics & small molecules, fondasi tetap penting untuk UC ringan–sedang, 5-ASA tetap lini awal–dan sulfasalazine (SSZ) masih relevan terutama bila cost/akses jadi isu dan tanpa kontraindikasi sulfa

SSZ = prodrug kolon-targeted: mengantarkan 5-ASA langsung ke mukosa kolon→ sehingga paling pas untuk UC dan selected colonic Crohn's (bukan ileal/perianal kompleks).

RCT & meta-analisis menunjukkan efikasi SSZ sebanding dengan mesalazine/5-ASA untuk induksi perbaikan dan maintenance remisi pada UC

tetap dipakai: treat-to-target → evaluasi early response (2–4 minggu) + biomarker (CRP/fecal calprotectin), lalu targetkan remisi klinis dan mucosal healing



Thank You